

# Optimal High-Frequency Market Making with Robust Backtesting

Garrett Naples, Dan Niedfeldt, Spencer Ozgur, Ben Zenker  
{gn2411, dn2648, so2770, bhz2110}@columbia.edu

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We focused on replicating and extending the results of two papers, the original Avellaneda-Stoikov paper and an adaptation by Stanford researchers: Takahiro Fushimi, Christian Gonzalez Rojas, and Molly Herman.

## 1 Background

### 1.1 Original A-S Paper

The original A-S paper derived the following model:

- Set an ‘indifference’ price of  $r = s - q\gamma\sigma^2(T - t)$ , where  $s$  is market mid,  $q$  is inventory size,  $\gamma$  is a risk-aversion parameter,  $\sigma$  is *standard deviation* of the stock price over the horizon  $T$ , and  $T - t$  is time from now ( $t$ ) until the horizon ( $T$ ).
- Set a bid-ask spread of  $\delta^a + \delta^b = \gamma\sigma^2(T - t) + \frac{2}{\gamma} \ln(1 + \frac{\gamma}{\kappa})$ .

In the derivation of this model, the following assumptions were made:

- The stock price follows arithmetic BM,  $dS_t = \sigma dW_t$ .
- The trader optimizes for expected CARA utility of wealth at time  $T$ ,
- The trader is allowed to liquidate inventory at time  $T$  at the mid-price.
- Order placements at a distance-to-mid of  $\delta$  are Poisson, with  $\lambda(\delta) = Ae^{-\kappa\delta}$ . More on  $\kappa$  later.
- The trader places orders of size 1.

In doing this, A-S found the following results:

- The distribution of PnL in numerical simulations had a lower mean than a baseline strategy of market making with a constant  $\delta$  around the mid, but had a much smaller standard deviation.
- Inventory sizes using this model were much smaller at EoD (and had a smaller standard deviation) than baseline.

Essentially, at the cost of some small amount of expected PnL, the A-S model was able to much more effectively control both risk ( $\sigma$  of PnL) and inventory.

### 1.2 Stanford Paper

The Stanford modification proposed/enacted the following changes:

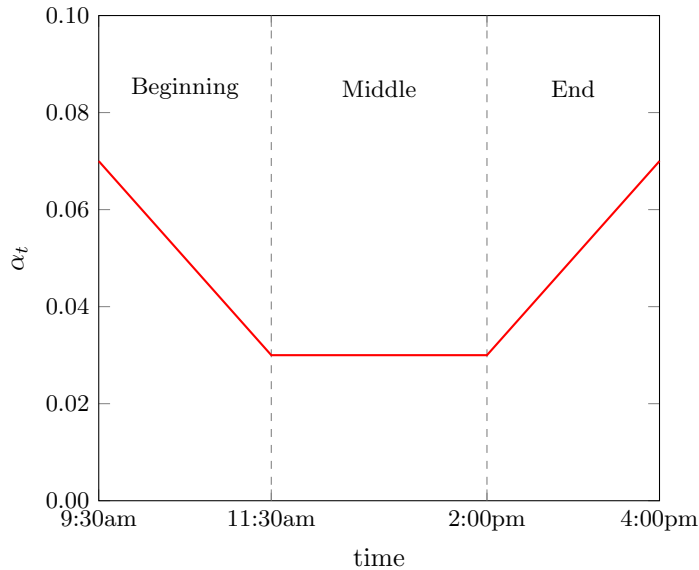
- Set order sizes through a proprietary sizing function. Let  $\varphi^{\max}$  be the maximum order size in shares, then set bid sizes according to the following functions:

$$\varphi_t^{\text{bid}} = \begin{cases} \varphi_t^{\max} & \text{if } q_t < 0 \\ \varphi_t^{\max} \cdot e^{-\eta q_t} & \text{if } q_t > 0 \end{cases}$$

$$\varphi_t^{\text{ask}} = \begin{cases} \varphi_t^{\max} & \text{if } q_t > 0 \\ \varphi_t^{\max} \cdot e^{-\eta q_t} & \text{if } q_t < 0 \end{cases}$$

where  $q_t$  is inventory at time  $t$ ,  $\varphi_t$  is the size of the order placed at time  $t$ , and  $\eta$  is a shape parameter.

- Add a time-varying modification to the Poisson parameter  $\lambda$ , where we now set  $\lambda(t, \delta) = \alpha_t e^{-\kappa\delta}$ .  $\alpha_t$  is a time-modifier, that scales  $\lambda$  up near the beginning and end of the trading day, and keeps it muted during the middle of the day, according to a piecewise-linear ‘bathtub’ model.



- Build a proprietary backtester, and test the A-S model with the proposed changes on real market data, as opposed to numerical simulations. They tested the 5 tickers AAPL, AMZN, GE, IVV, M, over the week 6/12/2017 - 6/16/2017.

Although A-S was built specifically for a controlled trading environment that obeys the econophysics of markets perfectly – this is nonetheless a useful thought experiment, to see which conclusions carry over into actual markets, and what other variables might need to be considered. They found similar conclusions: although it comes at a small cost of expected PnL, the ‘optimal’ strategy (with proprietary sizing, A-S indifference prices, and A-S spreads) has much better inventory control (a smaller standard deviation in EoD inventory). The market making algorithm in section 2.3 of the 2018 paper is given as follows:

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**Algorithm 1:** Market Making Algorithm

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while current time < end time do
  if no orders in the book then
    | Quote bid and ask prices
  else if 1 order in the book then
    | if current time - execution time > waiting time then
    |   | Cancel the outstanding order
    |   | Quote new bid and ask prices
    | else
    |   | Wait
    | end
  else if 2 orders in the book then
    | if current time - quote time > update time then
    |   | Cancel both order
    |   | Quote new bid and ask prices
    | else
    |   | Wait
    | end
  end
end
end

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## 2 Our Methodology/Improvements

Despite its many useful additions, the Stanford paper had a few flaws in its methodology that we utilized as areas for our own proprietary improvements/ideas.

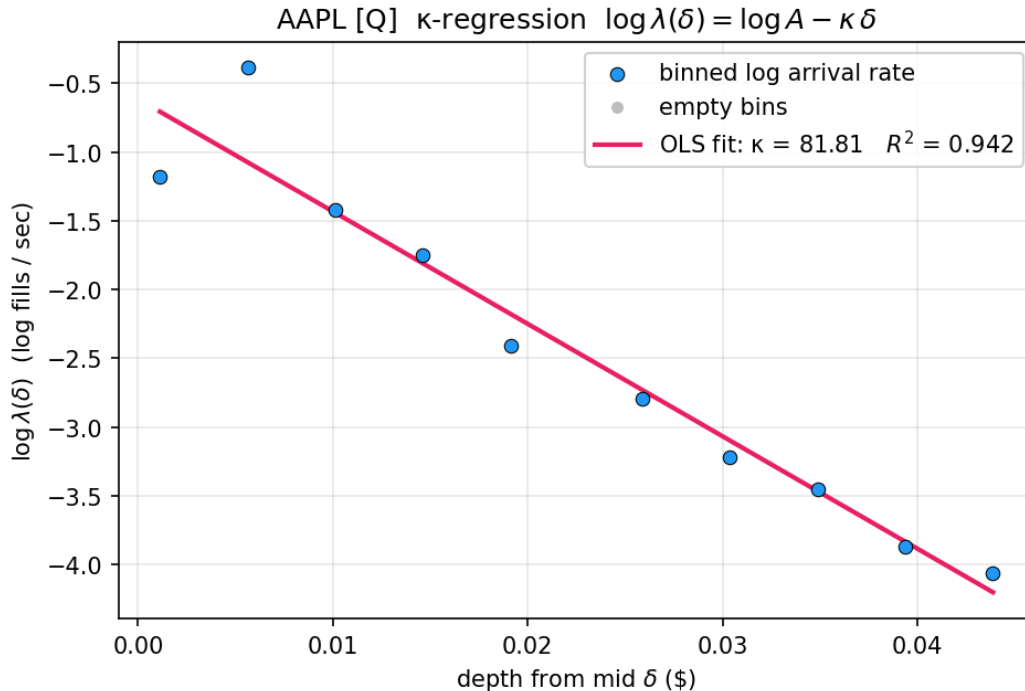
1. **Built Proprietary Backtester:** Despite being both a novel innovation, the Stanford authors’ backtester was relatively lackluster and lacked some methodological clarity. Our backtester used real WRDS market data (NBBO, trade tape) from the same timeframe, and mixed that with some assumptions about how the market behaves:
  - Used A-S assumption of Poisson order flow at a distance  $\delta$  from mid, for orders inside the NBBO.
  - Used historical market flow for orders on/outside the NBBO.
  - Used estimated values of  $\sigma$  based on previous week’s volatility, customized to individual tickers.
  - Assumed back-of-queue priority, which is worst-case scenario (and still didn’t do poorly!)
    - Built-in option to toggle queue priority between back and front

Again, we make some standard assumptions:

- No market impact
- No latency

- Single-exchange market making

2. **Calibrated  $\kappa$ :** We used exponential regression models to calibrate  $\kappa$  for each ticker, on real market data, where we regressed fill frequency on distance from mid. The Stanford paper uses a single unmotivated choice of  $\kappa$ . Pictured is the example linear regression performed on AAPL's trade data to obtain  $\kappa$ .



3. **True Parameter Estimation:** The Stanford paper, in order to more closely align with market data, essentially ‘cheats’ by calibrating  $\sigma$  and  $\gamma$  to real market participants. They set  $\sigma$  and  $\gamma$  for the course of the whole day, based on the open and close spreads, which are created by other participants. This artificially ensures their model can’t be far from truth. Instead, we were able to calibrate  $\sigma$  and  $\gamma$  fairly, and used reasonable values for each of these. We estimated  $\sigma$  historically, and used the commonly accepted  $\gamma = 0.01$  that was proposed in A-S.
4. **Enabled Rolling  $\sigma$  Usage:** We built out infrastructure to use a 10m rolling standard deviation as the  $\sigma$ -value, as opposed to daily  $\sigma$  estimated from the previous week. This allows quicker adaptation to changing market conditions, and allows our market maker to widen its spread based on recent volatility in the market. We decided not to use this for our results due to it being overly sensitive, but this can be an area for additional research.

### 3 Results

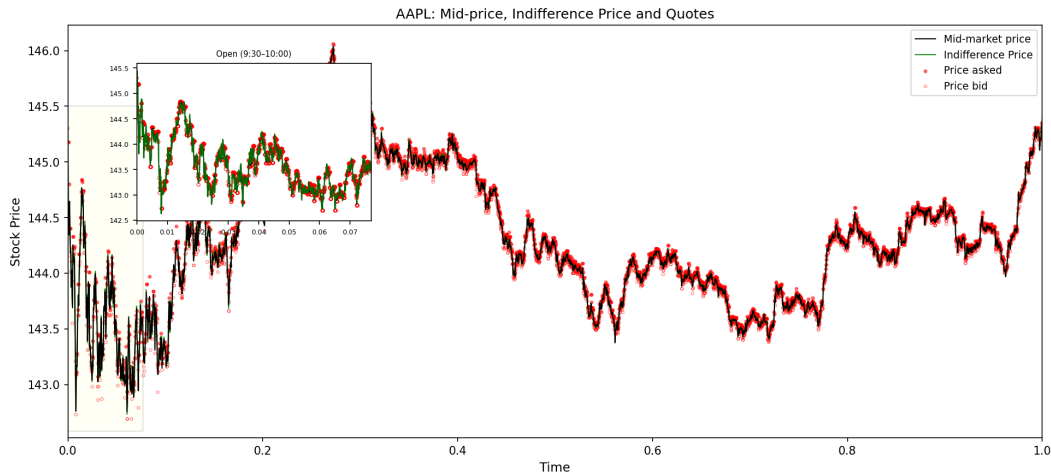
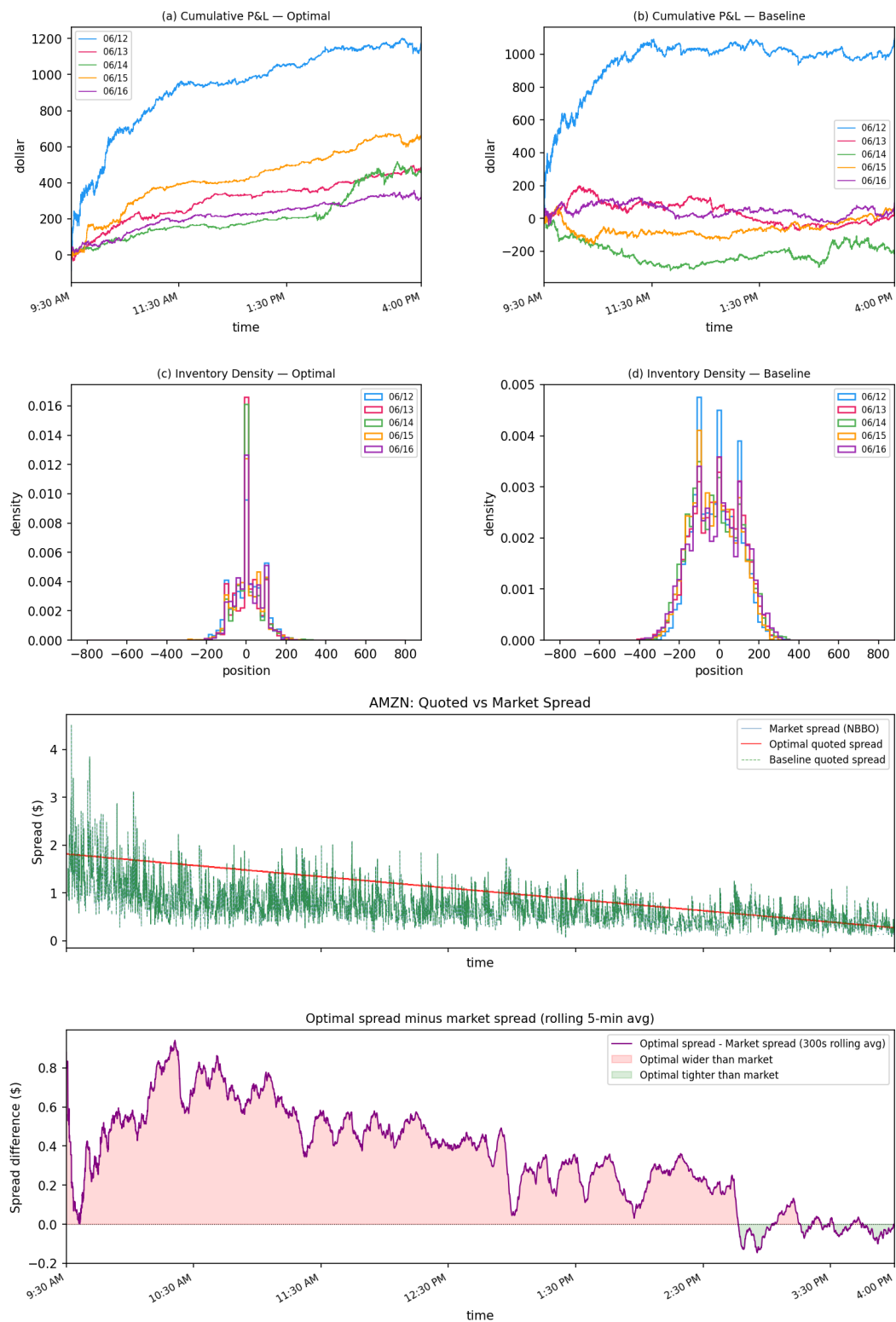


Figure 4: AAPL — P&L and Inventory



Included are some sample analysis charts. In short, our main takeaways are:

- Shockingly, our model was correctly calibrated to market open/close spread in most of our tickers. This was emphatically *not* a foregone conclusion: the spread only depends on  $\gamma, \sigma, \kappa$ , and we forced

ourselves not to tweak any of those parameters past the stated methods.  $\sigma$  was estimated using the previous week’s daily volatility,  $\kappa$  was estimated using linear (exponential) regression, and  $\gamma$  was set to a constant 0.01.

- The spread decreases linearly from market open to market close, as expected.
- The distribution of final inventory is *significantly* more concentrated around 0 when using optimal inventory control than it is for baseline. This is even more drastic of an improvement than seen in the original A-S paper, which is encouraging. The model aggressively flattens inventory positions by EoD, without much of a sacrifice of PnL. The optimal market making strategy actually improves on PnL over the baseline strategy for each of the 5 days for AAPL, and similarly shows marginal improvement in 3-4 days on the other symbols. This again is an improvement over both the A-S paper and the Stanford modification, where a slight drop in PnL was seen in the switch from baseline to ‘optimal’.

## 4 Future Work

While this project introduced a robust backtesting methodology to real world WRDS-TAQ (L1) data, future experiments could benefit from incorporating additional order book levels (L2) or Market-by-Order (L3) data from sources such as DataBento or LOBSTER. This would give access to direct queue position estimation and could allow for more sophisticated inventory control models using indicators like order book imbalance (OBI):  $(V_{\text{bid}} - V_{\text{ask}})/(V_{\text{bid}} + V_{\text{ask}})$ . Future work could also benefit from implementing a different trading algorithm from the one we outlined at the end of section 1.2., or by using a rolling window to calibrate parameters like  $\sigma$  and  $\kappa$  to allow the model to adapt to evolving market conditions as the trading day progresses.

## References

- [1] Avellaneda, M., & Stoikov, S. (2006). High-frequency trading in a limit order book.
- [2] Fushimi, T., Rojas, C. G., & Herman, M. (2018). Optimal High-Frequency Market Making.